

FretGarden

Pedagogical Constitution

Chapter 13

Interval, Scale-Degree, and Chord-Tone Pedagogy

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This chapter defines how FretGarden teaches the relational language that connects pitch, harmony, fretboard movement, and musical function.

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1. Purpose

This chapter defines standards for teaching intervals, scale degrees, chord tones, inversional relationships, and their use across hearing, naming, construction, navigation, performance, analysis, improvisation, and composition.

These domains form the relational core of FretGarden. They must be taught as connected capabilities rather than isolated vocabularies or static shapes.

2. Constitutional Authority

This chapter specializes the general constitutional requirements for objectives, evidence, lessons, exercises, assessment, mastery, review, feedback, remediation, sequencing, practice, and fretboard learning.

3. Governing Principle

Governing Rule: Relational concepts MUST be learned through coordinated sound, structure, location, function, and use. No single representation may stand in for the concept as a whole.

4. Core Distinctions

Interval number is not interval quality. Distance is not direction. Scale degree is not note name. Chord tone is not finger pattern. Root-relative function is not key-relative function. Recognition is not construction. Construction is not application.

5. Relational Domain Model

The domain includes melodic and harmonic intervals, simple and compound intervals, directed distance, scale-degree identity, tendency, chord membership, chord-tone function, inversion, guide tones, tensions, and voice-leading relationships.

6. Interval Foundations

Instruction should begin from perceivable and playable distance, then attach names and notation. Learners should compare same versus different, step versus leap, direction, size, and consonance or tension without treating subjective labels as definitions.

7. Number and Quality

Interval number derives from letter-name span; quality derives from semitone structure and diatonic context. Exercises must prevent learners from using semitone count alone when correct spelling is part of the objective.

8. Melodic and Harmonic Intervals

Melodic and harmonic presentations are distinct perceptual and performance conditions. Evidence in one condition does not automatically support mastery in the other.

9. Ascending and Descending Intervals

Direction changes perception, fingering, and melodic expectation. Core interval objectives should include both directions unless intentionally scoped.

10. Simple and Compound Intervals

Compound intervals should be connected to their simple-class relationships while preserving octave displacement and musical effect. Reduction is a reasoning tool, not a claim that compound intervals are identical in use.

11. Interval Construction

Construction tasks should vary roots, strings, directions, positions, notation, and response modes. The target interval must be necessary for success rather than inferable from a fixed shape color.

12. Interval Recognition

Recognition should include auditory, visual, notated, and fretboard forms. Selected-response items are useful for early discrimination but must progress toward naming, reproducing, and explaining.

13. Interval Fretboard Geometry

Shapes may encode interval classes, but learners must account for tuning relationships and the G-B string break. Multiple locations should be connected through note identity and function.

14. Scale-Degree Foundations

Scale degrees must be introduced as positions within a tonal collection relative to a tonic. Their names, numbers, sound, tendency, chord membership, and fretboard locations should be coordinated.

15. Movable-Do, Numbers, and Solfege

FretGarden may use numbers, scale-degree names, and movable-do syllables as complementary representations. One system should not be treated as universally superior; the selected representation must match the objective and remain consistently defined.

16. Tendency and Resolution

Tendency tones should be experienced through context and resolution, not memorized as absolute commands. Learners must distinguish stylistic expectation from mechanical inevitability.

17. Key and Mode Dependence

Scale-degree quality and tendency depend on collection, mode, harmony, and style. Major-key defaults must not be presented as universal properties of numbered degrees.

18. Chord-Tone Foundations

Chord tones must be learned as members of a chord relative to its root and quality. Root, third, fifth, seventh, extensions, alterations, and omissions should be connected to interval structure and sound.

19. Root, Bass, and Inversion

Root and bass are distinct. Inversion changes bass placement and voicing but does not change chord root. Exercises must directly test this distinction.

20. Guide Tones and Functional Priority

Thirds and sevenths often define quality and functional movement. Their importance should be taught through voice leading and harmonic context rather than as a universal ranking detached from style.

21. Chord Tones Versus Scale Degrees

A pitch may simultaneously have a scale-degree function and a chord-tone function. Learners must practice switching frames and explaining which reference point is active.

22. Bidirectional Mapping

Core work must map names to sounds, sounds to names, functions to fretboard locations, locations to functions, symbols to constructions, and musical contexts to intentional choices.

23. Representation Coordination

Staff notation, tablature, fretboard diagrams, interval labels, scale-degree numbers, solfege, chord symbols, and audio should reinforce rather than contradict one another. Redundant representations should be removed when they reduce active retrieval.

24. Ear Integration

Learners should sing, hear, compare, and reproduce relational content. Singing accuracy may be interpreted separately from conceptual identification where vocal control is not the target.

25. Rhythm and Temporal Context

Intervals and chord tones should eventually be produced in time, over changing harmony, and within phrases. Static untimed accuracy is necessary but insufficient for applied fluency.

26. Improvisational Application

Application tasks should constrain target intervals, degrees, or chord tones while preserving musical choice. Learners may target guide tones, resolve tendencies, create motifs, or vary one relationship across chords.

27. Compositional Application

Composition tasks should require deliberate use and explanation of relational choices. Evaluation must allow multiple valid outcomes while checking whether the target relationship is present and intentional.

28. Analysis

Analysis should connect labels to audible and performable consequences. Naming every interval or chord tone without explaining function or use is incomplete when the objective is musical reasoning.

29. Sequencing

Intervals should support scale-degree and chord-tone learning, but the curriculum may spiral these domains rather than complete one entirely first. Prerequisites must be objective-specific.

30. Assessment

Assessment should sample varied roots, keys, directions, textures, positions, and representations. Strong claims require constructed or performed responses, delayed retrieval, and transfer beyond practiced examples.

31. Mastery

Mastery profiles should distinguish naming, hearing, constructing, locating, explaining, and applying. A learner may be strong in one dimension and emerging in another.

32. Review

Review should interleave confusable intervals, vary tonal contexts, rotate harmonic functions, and require retrieval before re-exposure. Familiar songs may support memory but cannot be the sole cue.

33. Error Classification

Errors may involve counting, quality, spelling, direction, root reference, tonic reference, inversion, chord membership, tuning-break geometry, auditory confusion, or context selection.

34. Remediation

Remediation should isolate the failed relation, contrast it with a confusable neighbor, restore an anchor, and then return to varied application. Answer revelation must be followed by unsupported reconstruction.

35. Adaptive Difficulty

Variables include interval set, key, direction, register, harmonic density, tempo, response mode, number of simultaneous reference frames, and degree of context.

36. Accessibility

Equivalent response modes may separate hearing, conceptual reasoning, and physical performance. Color, notation, audio, and motor demands must not be conflated with relational knowledge.

37. Anti-Patterns

Noncompliant patterns include memorizing song mnemonics without transfer, teaching scale degrees as fixed notes, treating chord tones as colored dots, ignoring root versus bass, assessing only major-key examples, and using one fretboard shape as complete evidence.

38. Constitutional Non-Negotiables

1. Interval, scale-degree, and chord-tone objectives MUST identify the active reference point.
2. Core instruction MUST connect sound, structure, and use.
3. Shape knowledge MUST be unpacked into interval or note content.
4. Broad claims MUST include varied roots and contexts.
5. Root, tonic, and bass MUST remain distinct.
6. Auditory and physical evidence MUST be reported according to their actual conditions.
7. Application MUST include meaningful learner choice.
8. Review MUST include confusable alternatives and delayed retrieval.

39. Conformance Tests

Reviewers must identify the reference frame, target relation, representation, required response, evidence limits, transfer condition, common confusions, review plan, and musical application.

40. Tradeoffs and Resolution Rules

Terminological precision versus beginner load: introduce only distinctions needed now, but do not teach false simplifications.

Shape efficiency versus relational understanding: use shapes as scaffolds and require reconstruction.

Functional labels versus note names: coordinate both and vary the requested frame.

41. Assumptions Made

This chapter assumes tonal and modal contexts as primary entry points while allowing later expansion into chromatic, post-tonal, and style-specific systems.

42. Unresolved Questions

Unresolved matters include default interval sequences, solfege policy, auditory threshold calibration, compound-interval milestones, style-specific tendency models, and automated evaluation of sung or played responses.

43. Topics Intentionally Deferred

Advanced chromatic harmony, set theory, microtonality, species counterpoint, jazz-tension taxonomies, and exhaustive orchestration are deferred.

44. Required Future Cross-References

Rhythm, ear training, harmony, improvisation, composition, analytics, accessibility, and quality-assurance specifications must preserve the reference-frame distinctions established here.

45. Chapter Conformance Statement

A compliant system develops relational understanding that can be heard, named, constructed, located, explained, retained, transferred, and used musically without dependence on one cue or pattern.